

Applied Probability and Statistics

DA-IICT

Course Syllabus

Course Title

Applied Probability and Statistics

Credit Structure

Lecture hours per week: 3

Tutorial hours per week: 0

Practical hours per week: 0

Total Credits: 3

Program/Semester

MTech 1st Sem, MSc DS 1st Sem, BTech 7th Sem

Prerequisites Courses

Calculus

Instructor

Dr. Abhishek Tilva

Course Description

This course provides a rigorous introduction to probability and its applications in stochastic modelling, statistics and data science. Many applied topics such as martingales, stochastic processes, regression, etc will be discussed.

Course Objectives and Outcomes

This course is designed to equip students with a rigorous understanding of probability theory and its applications in modeling uncertainty across real-world phenomena. The course aims to build students' ability to formulate, analyze, and implement probabilistic and statistical models relevant to domains such as data science, finance, and engineering.

Upon completion of this course, students will be able to:

- Develop understanding of concepts in probability and statistics and related notions such as stochastic processes.
- Utilize probabilistic reasoning and modeling tools in research and industry applications across fields such as finance, data science, operations research, and engineering.
- Apply statistical techniques such as regression and statistical design to model and interpret real-world data.

Course Materials/References

Suggested Textbooks

- Grimmett, G. R., and D. R. Sterzaker. "Probability and Random Processes" 4th edition."
- DeGroot, Morris H., and Mark J. Schervish. "Probability and statistics" 4th edition."
- Ramsey, Fred and Schafer, Daniel. "The Statistical Sleuth: A course in methods of Data Analysis" 3rd edition.
- F. E. Harrell, Jr. "Regression Modelling Strategies with Applications to Linear Models, Logistic Regression and Survival analysis."

Evaluation/Grading Policy

- Midterm: 40%
- Final: 60%

Detailed Course Contents**

** The contents are subject to change based on time and other factors.

Topic Name	Content (2-3 lines per 4-6 lectures)	No. of Lectures
Foundations of probability	Probability spaces, σ -fields, conditional probability, independence, random variables and vectors, Borel-Cantelli Lemmas	3
Discrete and continuous random variables	Cdf, Pmf, Pdf, moments, common discrete and continuous distributions, conditional distributions and expectation, functions of random variables	4
Generating functions and applications	Different types of generating functions, branching processes, convergence and asymptotics, large deviations	3
Martingales	Martingales, sub and supermartingales, optional stopping, random walk, wald's identity	5
Stochastic processes	Brownian motion, Gaussian processes, Stationary and ergodic processes, Stochastic calculus and applications	6
Statistical inference	Maximum likelihood estimation, Bayesian inference, Hypothesis testing, Confidence intervals	6
Regression and machine learning	Linear regression, logistic regression, generalized linear models (GLMs), supervised and unsupervised learning basics	6
Case studies and design of experiments	Inference based on t-distribution, ANOVA, Categorical data analysis and non-parametric methods	6
Sampling and bootstrap	Inverse transform, rejection sampling, importance sampling, MCMC and Gibbs sampling, parametric and non parametric bootstrap	5
Miscellaneous topics	Advanced data analysis, reliability theory, etc	6